

Fan Jiang

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Educations

The University of Melbourne

PhD in Natural Language Processing

- Supervisors: Prof. Trevor Cohn & Prof. Tom Drummond

Melbourne, Victoria, Australia

Feb. 2022 - Oct. 2025

The University of Melbourne

Master of Science (Computer Science) (with Distinction)

- GPA: 91.9% (3.94/4.0) First Class Honours
- Supervisor: Prof. Trevor Cohn

Melbourne, Victoria, Australia

Jul. 2019 - Jun. 2021

Fuzhou University

Bachelor of Engineering in Software Engineering

- GPA: 4.07/5.0 (Ranking: 1/155) Graduated with Honours

Fuzhou, Fujian, China

Sep. 2015 - Jun. 2019

Research Interest

My primary research interests are centered around natural language processing. My research is motivated by the **ultimate goal of enhancing information access for individuals across diverse backgrounds**. Recently, I have been concentrating on achieving robust and general information access by overcoming linguistic and domain barriers:

- General knowledge retrieval systems across domains** – Robust systems capable of retrieving reliable information across various formats and domains are crucial for meeting the information requirements of intricate real-world user queries across diverse fields.
- Accessing information across languages** – It is crucial to enhance information accessibility for users from diverse linguistic backgrounds, particularly those who speak underrepresented languages. My emphasis is on developing dedicated systems capable of aiding linguistic minorities in accessing essential information.

Publications

Preprints	Marco-MoE: Open Multilingual Mixture-of-Expert Language Models with Efficient Upcycling. Fan Jiang , Yu Zhao, Chenyang Lyu, Tianqi Shi, Yichao Du, Feihu Jiang, Longyue Wang, Weihua Luo Under Review.
Preprints	Difficulty-Estimated Policy Optimization. Yu Zhao*, Fan Jiang* , Tianle Liu*, Bo Zeng, Yu Liu, Longyue Wang, Weihua Luo arXiv Preprint 2602.06375
EACL'26	Tokenizer-Aware Cross-Lingual Adaptation of Decoder-Only LLMs through Embedding Relearning and Swapping. Fan Jiang , Honglin Yu, Grace Chung and Trevor Cohn. In Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics.
TACL'25	Few-Shot Multilingual Open-Domain QA from Five Examples. Fan Jiang , Tom Drummond and Trevor Cohn. Transactions of the Association for Computational Linguistics, 13:481–504.
MRL'24	Language Bias in Multilingual Information Retrieval: The Nature of the Beast and Mitigation Methods. Jinrui Yang, Fan Jiang , and Timothy Baldwin. In Proceedings of the Fourth Workshop on Multilingual Representation Learning.
EMNLP'24	Pre-training Cross-lingual Open Domain Question Answering with Large-scale Synthetic Supervision. Fan Jiang , Tom Drummond and Trevor Cohn. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing.

EMNLP'23	Boot and Switch: Alternating Distillation for Zero-Shot Dense Retrieval. Fan Jiang , Qiongkai Xu, Tom Drummond and Trevor Cohn. In Findings of the Association for Computational Linguistics: EMNLP 2023.
EMNLP'23	Noisy Self-Training with Synthetic Queries for Dense Retrieval. Fan Jiang , Tom Drummond and Trevor Cohn. In Findings of the Association for Computational Linguistics: EMNLP 2023.
EACL'23	Don't Mess with Mister-in-Between: Improved Negative Search for Knowledge Graph Completion. Fan Jiang , Tom Drummond and Trevor Cohn. Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics.
AAAI'22	Incorporating Constituent Syntax for Coreference Resolution. Fan Jiang and Trevor Cohn. Proceedings of the 36th AAAI Conference on Artificial Intelligence.
NAACL'21	Incorporating Syntax and Semantics in Coreference Resolution with Heterogeneous Graph Attention Network. Fan Jiang and Trevor Cohn. Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies.

Work Experience

Alibaba Group	<i>Hangzhou, China</i>
Senior Researcher	<i>Sep. 2025 - Now</i>
Full Time Employee	Manager: Longyue Wang
Multilingual MoE LLMs with compact sizes: Architected highly-sparse Mixture-of-Experts models utilising efficient upcycling to optimise localised knowledge understanding; disclosed full training recipes, datasets, and weights to the research community. Efficiency in reasoning RL: Developed a novel online Difficulty Estimator to dynamically filter training data during the rollout phase, prioritising computational resources for high-learning-potential samples. Multicultural reasoning benchmark: Co-designed a rigorous multilingual benchmark that established a new industry standard for cultural reasoning, where current SOTA models (e.g., Gemini 2.5 Pro) achieve only 18.5% accuracy on the most difficult subsets.	

Research Experience

Google Research Australia	<i>Sydney, NSW, Australia</i>
Project: Franken-Adapter: Cross-Lingual Adaptation of LLMs by Embedding Surgery.	<i>Jun. 2024 - Dec. 2024</i>
Student Researcher	Hosts: Trevor Cohn & Honglin Yu
- Developed a modular framework for efficient cross-lingual adaptation, enabling LLMs with size up to 35B to support 100+ languages through specialised tokenizer customization and embedding layer "surgery". - Engineered a model composition approach to achieve zero-shot cross-lingual transfer, seamlessly integrating customized embedding layers into post-trained LLM backbones. - Optimised training efficiency by isolating embedding re-learning, significantly reducing the computational overhead compared to full-parameter fine-tuning for achieving language adaptation of LLMs.	
The University of Melbourne	<i>Melbourne, Victoria, Australia</i>
Ph.D. Thesis: Efficient Self-Supervised Adaptation for Information Access Across Low-Resource Languages and Domains	<i>Mar. 2022 - Aug. 2025</i>
PhD Student	Advisors: Prof. Trevor Cohn & Prof. Tom Drummond
- Structured knowledge: adapting neural retrievers to knowledge graphs. (EACL 2023) - Domain adaptation: adapting neural retrievers to zero-resource domains with a novel self-training approach. (EMNLP 2023) - General neural retriever: a bootstrapping algorithm for unsupervised retriever training across massive domains. (EMNLP 2023) - A unified system for cross-lingual retrieval and multilingual ODQA: a two-stage self-supervised pre-training framework with large-scale synthetic data for strong zero-shot cross-lingual retrieval and multilingual QA performance. (EMNLP 2024) - Few-shot multilingual ODQA: a data generation approach that employs LLMs for generating synthetic data from 5-shot in-language examples, and can be extended to zero-shot setting to enable adaption to low-resource languages with only English supervision. (TACL 2025)	

